Celestial Coordinate Systems: Alt–Az and RA–Dec

What you will be able to do

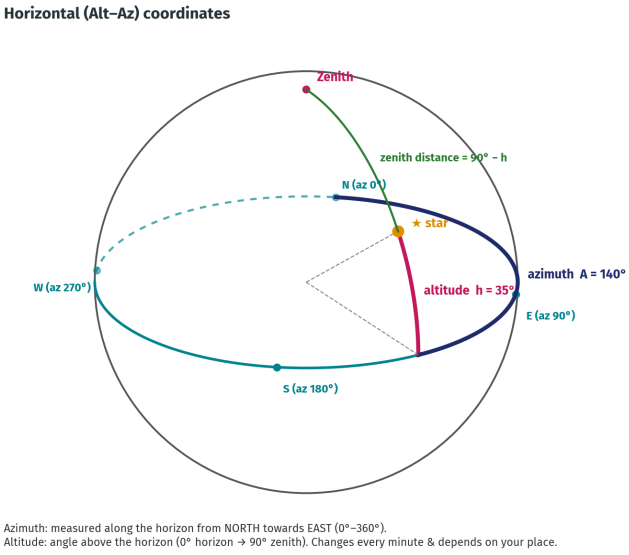
1. Describe a position in the sky with **altitude and azimuth** — and explain why these change.
2. Describe a position with **right ascension (RA) and declination (Dec)** — and explain why these stay fixed.
3. Convert RA between hours and degrees.
4. Calculate a star's **altitude when it crosses the meridian**.
5. Use **local sidereal time (LST)** to find which stars are on the meridian at a given time.

1. Why we need coordinates

"The bright star over that tree" is no help to a friend in another city or on another night. Like latitude and longitude on Earth, the sky needs a grid:

System	Reference circle	Zero point	Changes with time & place?	Best for
Horizontal (Alt–Az)	your horizon	North point	**yes**, every minute	"where do I look *now*?"; Dobsonian telescopes
Equatorial (RA–Dec)	celestial equator	First Point of Aries	**no** (only very slowly)	star catalogues, maps, equatorial mounts

2. The horizontal system: altitude and azimuth



Altitude is measured up from the horizon; azimuth along the horizon from North towards East.

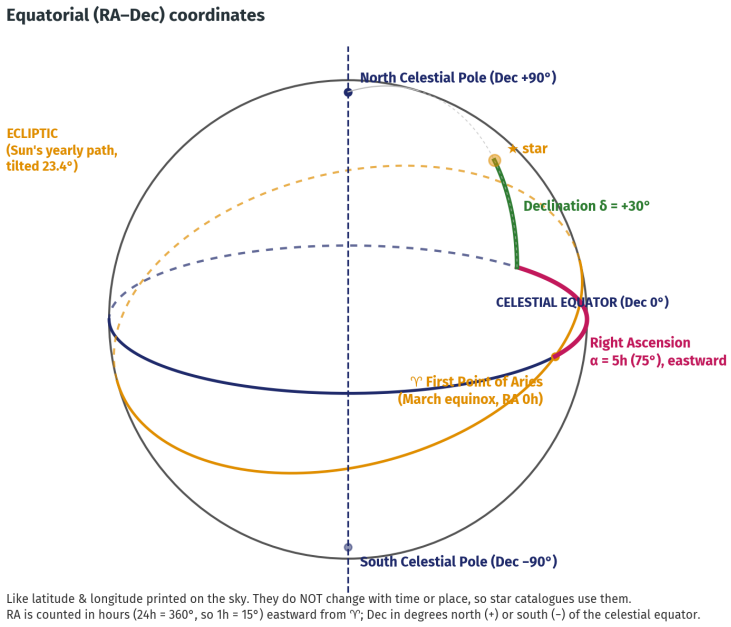
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P.T.O.

- **Altitude** h: angle **above the horizon**, 0° at the horizon, 90° at the zenith (negative = below the horizon).
- **Azimuth** A: angle **along the horizon from North towards East**: N = 0°, E = 90°, S = 180°, W = 270°.
- **Zenith distance** $z = 90° - h$.

A star at $A = 135°$, $h = 30°$ is in the **south-east**, three fists up. **Drawback:** as Earth turns, every star's altitude and azimuth change constantly — and differ between Mumbai and Delhi at the same instant.

3. The equatorial system: right ascension and declination



RA is counted eastward along the celestial equator from the First Point of Aries; Dec is measured north or south of the equator.

Earth	Sky
Latitude (N/S of the equator)	**Declination** δ: +90° (NCP) to −90° (SCP)
Longitude (E/W of Greenwich)	**Right ascension** α: measured **eastward** from the First Point of Aries, 0^h to 24^h

Why hours?

The sky turns 360° in about 24 hours, so it is natural to measure RA in time units:

$$24^h = 360° \;\Rightarrow\; 1^h = 15°, \quad 1^m = 15', \quad 1^s = 15''$$

Worked examples

1. Betelgeuse, $\alpha = 5^h55^m$: $5 \times 15° + 55 \times \frac{15}{60}° = 75° + 13.75° = \mathbf{88.75°}$.

2. An object at $\alpha = 202.5°$: $202.5/15 = 13.5^h = \mathbf{13^h30^m}$.

RA and Dec are attached to the stars, so they are the **same for everyone, everywhere, every night** — a catalogue can list each star once.

Coordinates of some bright stars (epoch J2000)

Star	Constellation	RA	Dec	Indian name
Sirius	Canis Major	6^h45^m	$-16°43'$	Lubdhaka
Canopus	Carina	6^h24^m	$-52°42'$	Agastya
Arcturus	Boötes	14^h16^m	$+19°11'$	Swati
Vega	Lyra	18^h37^m	$+38°47'$	Abhijit
Capella	Auriga	5^h17^m	$+46°00'$	Brahmahridaya
Betelgeuse	Orion	5^h55^m	$+7°24'$	Ardra
Aldebaran	Taurus	4^h36^m	$+16°31'$	Rohini
Antares	Scorpius	16^h29^m	$-26°26'$	Jyeshtha
Spica	Virgo	13^h25^m	$-11°10'$	Chitra
Altair	Aquila	19^h51^m	$+8°52'$	Shravana
Regulus	Leo	10^h08^m	$+11°58'$	Magha
Polaris	Ursa Minor	2^h32^m	$+89°16'$	Dhruva
Pleiades (Alcyone)	Taurus	3^h47^m	$+24°06'$	Krittika

Epoch: because of precession (Lecture 1.2), RA and Dec drift slowly (about 50″ a year), so catalogues state the date they refer to — usually **J2000.0**.

4. The meridian: when is a star highest?

A star is highest when it crosses your meridian (its *upper culmination* or *transit*):

$$\boxed{h_{\text{transit}} = 90° - \varphi + \delta} \qquad (\text{if} > 90°\text{: north of the zenith, at } 180° - h)$$

This is the same formula as for the noon Sun in Lecture 1.2 — the Sun is just another star with a declination!

For a circumpolar star the **lower culmination** (passing below the pole) is at $h_{\text{lower}} = \varphi + \delta - 90°$.

Worked examples

1. Sirius from Mumbai: $90° - 19.1° - 16.7° = \mathbf{54.2}°$, due south (about 5½ fists).

2. Canopus from Delhi: $90° - 28.6° - 52.7° = \mathbf{8.7}°$ — just above the southern horizon.

3. Kochab ($\delta = +74.2°$) from Delhi: upper $= 90 - 28.6 + 74.2 = 135.6° > 90° \Rightarrow 180 - 135.6 = \mathbf{44.4}°$ (north); lower $= 28.6 + 74.2 - 90 = \mathbf{12.8}°$ — still above the horizon, so circumpolar ✔

Stars that pass through your zenith

A star passes **exactly overhead** when $\delta = \varphi$:

City	Latitude	Bright star passing (nearly) overhead
Delhi	28.6° N	Pollux (+28.0°), Elnath (+28.6°)
Kolkata	22.6° N	the Pleiades (+24°) pass within 1.5°
Mumbai	19.1° N	**Arcturus** (+19.2°) — almost exactly overhead!
Bengaluru / Chennai	13° N	Regulus (+12.0°)

5. Hour angle and local sidereal time

- **Local Sidereal Time (LST)** = the RA that is on your meridian right now — a "star clock".
- **Hour angle** of a star: $\mathrm{HA} = \mathrm{LST} - \alpha$ (positive: already past the meridian, in the west; negative: still in the east).

A star is on the meridian when LST = its RA.

Estimating LST (accurate to about 15 minutes)

At local noon the Sun is on the meridian, so LST ≈ the Sun's RA, which grows by about 2 hours per month:

Date	21 Mar	21 Apr	21 May	21 Jun	21 Jul	21 Aug	23 Sep	23 Oct	22 Nov	22 Dec	20 Jan	19 Feb
Sun's RA	0^h	2^h	4^h	6^h	8^h	10^h	12^h	14^h	16^h	18^h	20^h	22^h

$$\boxed{\mathrm{LST} \approx \alpha_\odot + (\text{local time} - 12^h)} \qquad \text{local mean time} = \mathrm{IST} + 4\ \text{min} \times (\lambda - 82.5^\circ)$$

(add or subtract 24^h if needed).

Worked examples

1. What is on the meridian at 9 PM IST on 20 January in Nagpur (79.1° E)?
Local time = 21:00+4×(79.1−82.5) min = 20:46 = 20.8^h; LST ≈ 20+(20.8−12) = 28.8−24 = $\mathbf{4.8^h}$. So stars with RA ≈ 4^h48^m are on the meridian: **Aldebaran** (4^h36^m) has just crossed, almost overhead ($90 - 21.1 + 16.5 = 85.4^\circ$).

2. In which month is Antares ($\alpha = 16.5^h$) on the meridian at 9 PM? $16.5 = \alpha_\odot + 9 \Rightarrow \alpha_\odot = 7.5^h \Rightarrow$ **mid-July** — Scorpius is a monsoon-season constellation.

6. Rising and setting: quick rules

Condition (northern observer)	Result
$\delta > 90^\circ - \varphi$	circumpolar (always up)
$\delta < -(90^\circ - \varphi)$	never rises
$\delta = 0^\circ$	rises due east, sets due west, up for exactly 12 hours
$\delta > 0^\circ$	rises north of east; up for more than 12 hours
$\delta < 0^\circ$	rises south of east; up for less than 12 hours

7. Other systems (good to know)

- **Ecliptic coordinates**: longitude λ along the ecliptic from the First Point of Aries, and latitude β. The positions in a Panchang are (sidereal) ecliptic longitudes: "Moon in Rohini" means the Moon's longitude lies in that $13^\circ 20'$ span.
- **Galactic coordinates**: centred on the centre of our Milky Way, which lies in Sagittarius (RA 17^h46^m, Dec -29°).

Try tonight

1. In Stellarium (free), click a bright star and note its Az/Alt and RA/Dec. Wait 15 minutes and check again — which numbers changed?
2. Estimate the same star's altitude with your fist and compare.
3. Find the LST in Stellarium's object information and check that the star with RA $\approx$ LST is due south.

Key points to remember

- Alt–Az: azimuth from North through East; altitude above the horizon. Changes constantly and with place.
- RA–Dec: RA in hours eastward ($1^{\mathrm{h}} = 15^\circ$); Dec in degrees $\pm$. Fixed to the stars.
- $h_{\mathrm{transit}} = 90^\circ - \varphi + \delta$; overhead when $\delta = \varphi$.
- On the meridian when LST = RA; LST $\approx \alpha_\odot +$ (local time $- 12\,\mathrm{h}$), with $\alpha_\odot$ growing 2 h per month from 0 h on 21 March.

Check yourself

1. Convert RA $18^{\mathrm{h}}37^{\mathrm{m}}$ (Vega) into degrees.
2. What is Vega's transit altitude from Bengaluru (13° N), and in which direction?
3. Is Canopus visible from Srinagar (34.1° N)?
4. At about what local time is Sirius ($6^{\mathrm{h}}45^{\mathrm{m}}$) on the meridian on 22 December?

Answers: 1. $18\times15+37\times0.25 = 279.25^\circ$. 2. $90-13+38.8 = 115.8^\circ \Rightarrow 64.2^\circ$, north of the zenith. 3. $90-34.1-52.7 = 3.2^\circ$: only just, lost in haze. 4. $6.75 = 18 + (t - 12) \Rightarrow t \approx 0.75^{\mathrm{h}}$, about 12:45 AM.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §4.1 "Earth and Sky" (latitude, longitude, celestial coordinates) — https://openstax.org/books/astronomy-2e/pages/4-1-earth-and-sky
- Junior IOAA syllabus, "Coordinates and Times" — https://ioaastrophysics.org/junior-ioaa/junior-ioaa-syllabus (for olympiad aspirants)